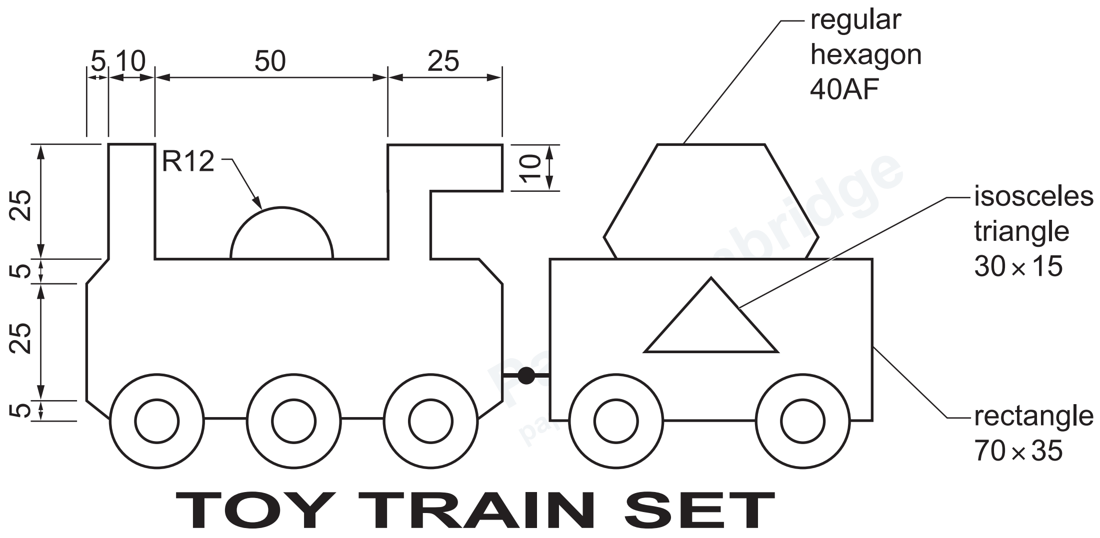
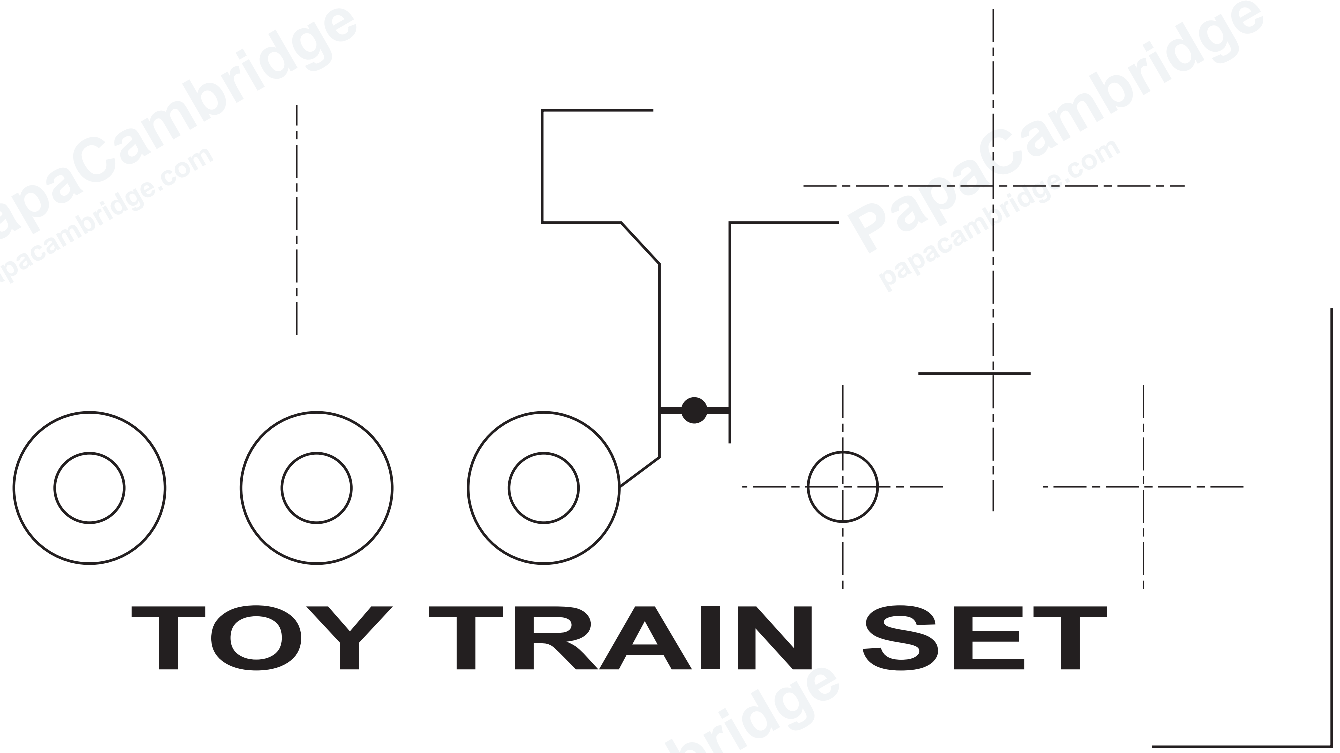
Section A

Answer **all** questions in this section.

A1 A design for the lid of a toy train set box is shown below.





(a) Complete the full-size drawing of the design for the train set box lid by adding:

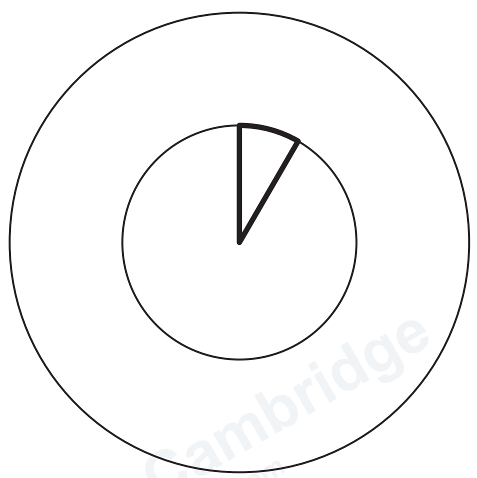
- **(i)** the train outline [4]
- **(ii)** the wheels [2]
- **(iii)** the rectangle [2]
- **(iv)** the triangle [2]
- **(v)** the hexagon. [3]

(b) Complete the 190 mm × 105 mm rectangular border to the design for the train set box lid. [2]

A2 The wheels on the drawing of the toy train are to have 12 evenly spaced segments.

One segment has already been added to the wheel.

Complete the drawing of the toy train wheel by adding the missing 11 evenly spaced segments. [3]



toy train wheel

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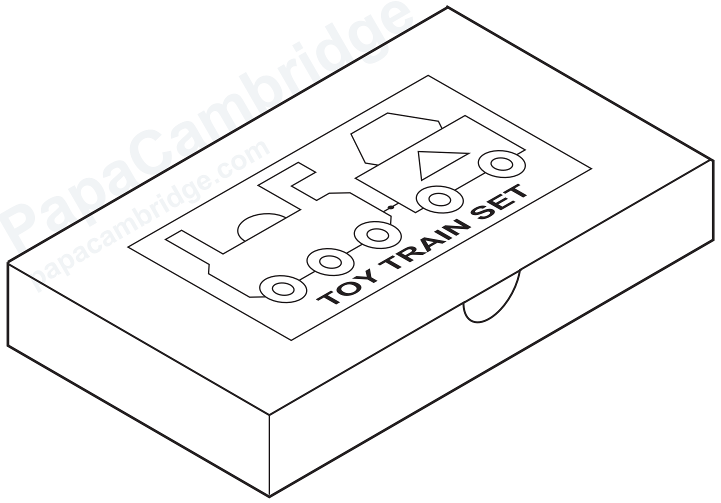
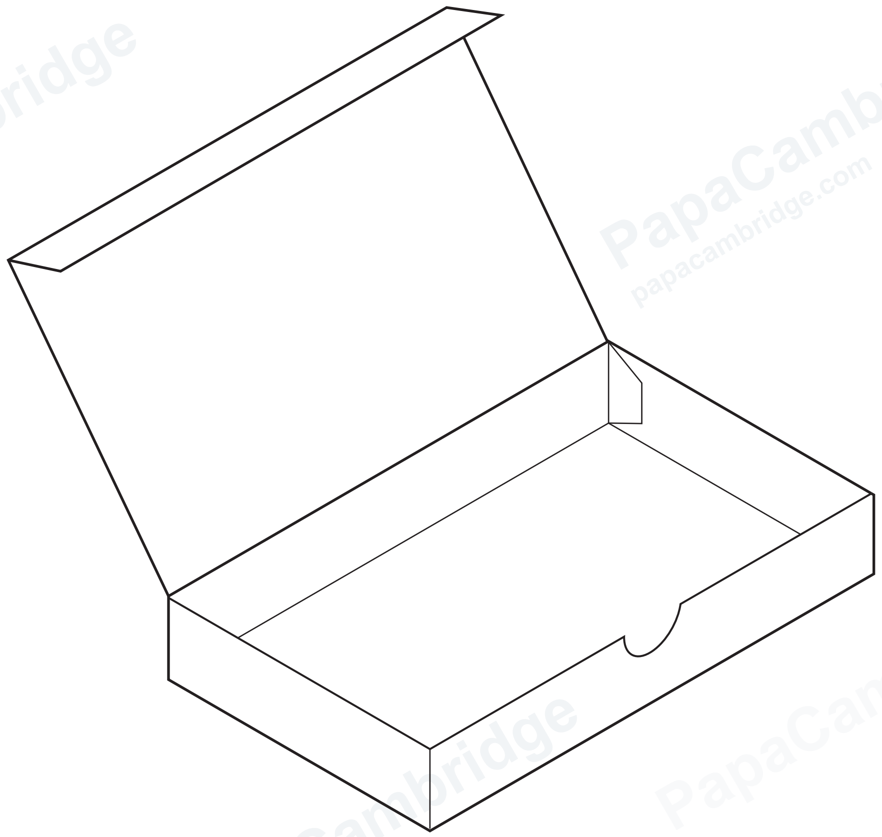
A3 The train set box is shown below.

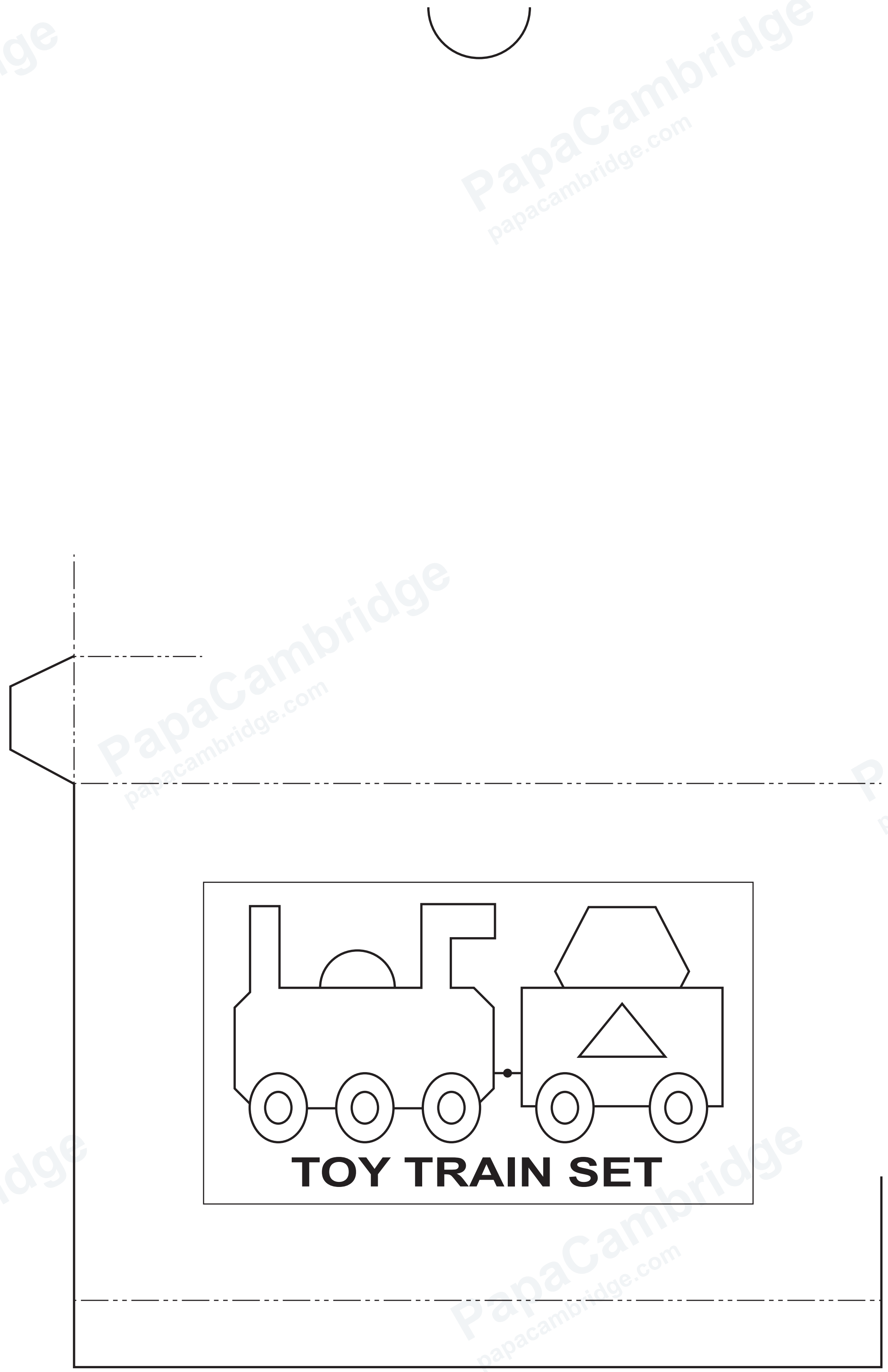
The box is made from thin card.



closed box



open box



development (net)

Complete the development (net) of the train set box to a scale of 1 : 2. [7]

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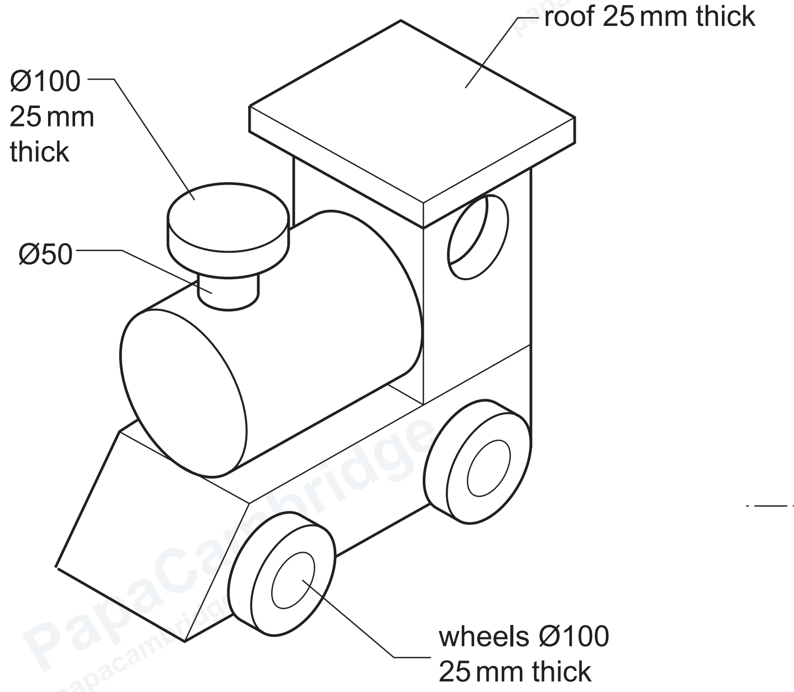
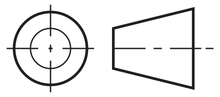
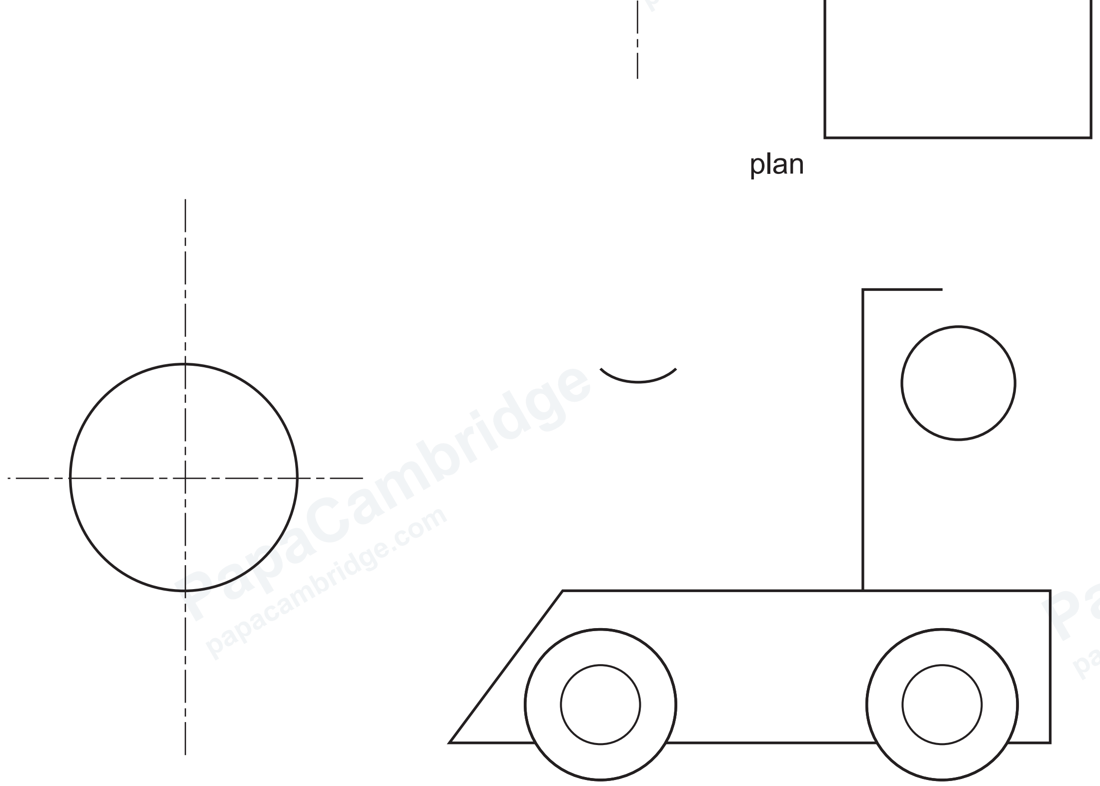
Section B

Answer **one** question, **either** Question **B4** or **B5**, from this section.

B4 A toy train is shown below.





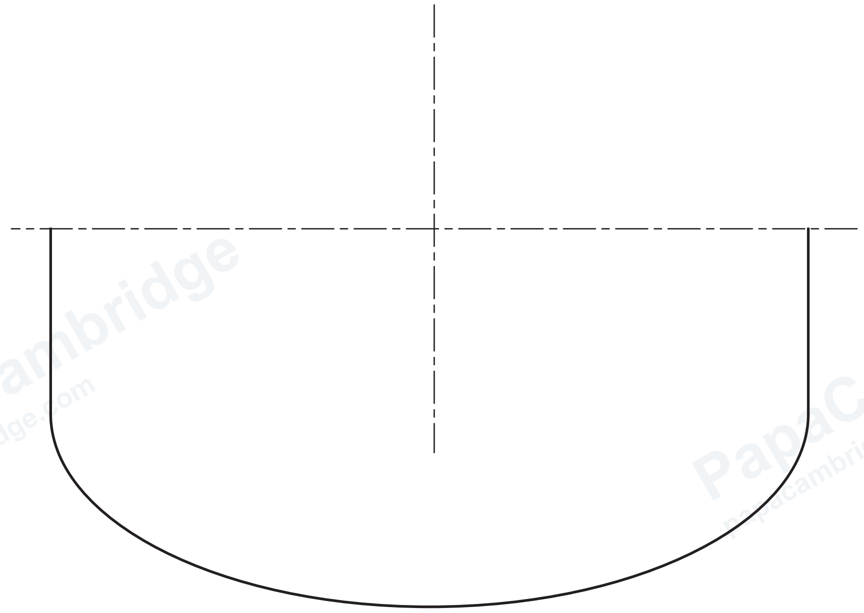


(a) Complete the orthographic views of the toy train to a scale of 1 : 5. [13]

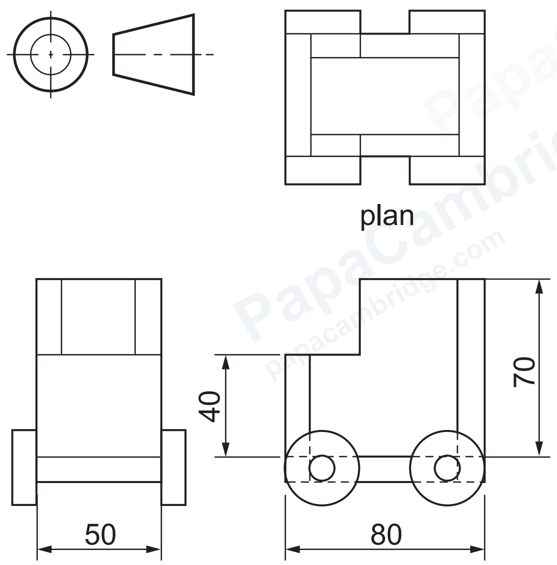
Ignore any hidden detail.

(b) Part of the chimney on the toy train is shown below.

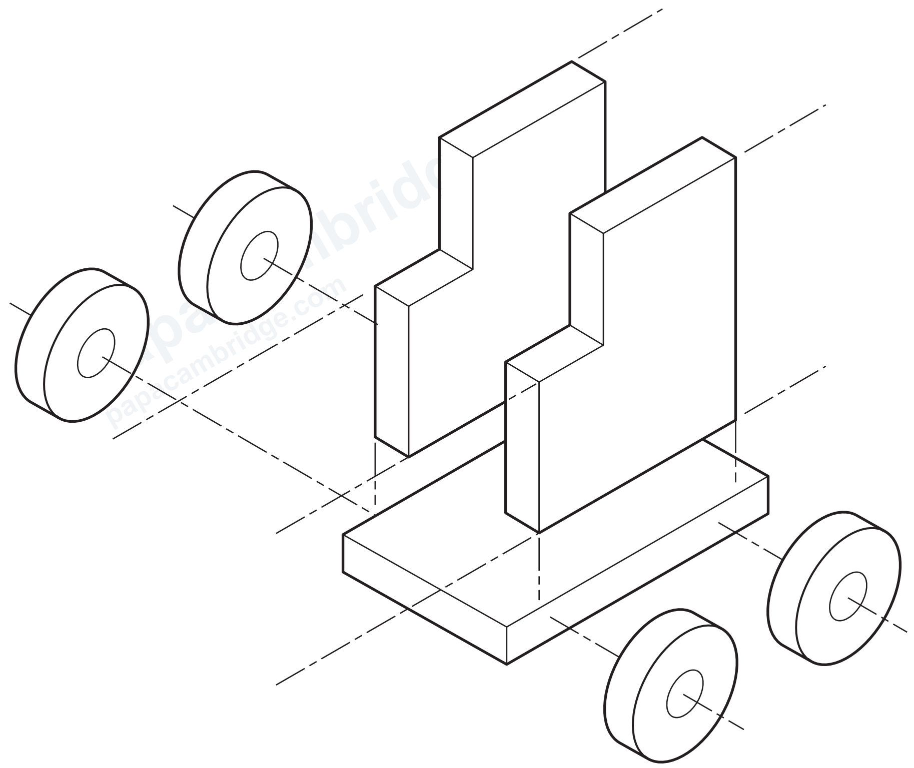
Complete the top section of the chimney by adding the ellipse.
Major axis 100. Minor axis 50. [6]



(c) Orthographic views of a truck for the toy train are shown below.
The truck is made from 10 mm thick softwood.



Complete the exploded isometric view of the truck to a scale of 1 : 2. [6]


exploded isometric view

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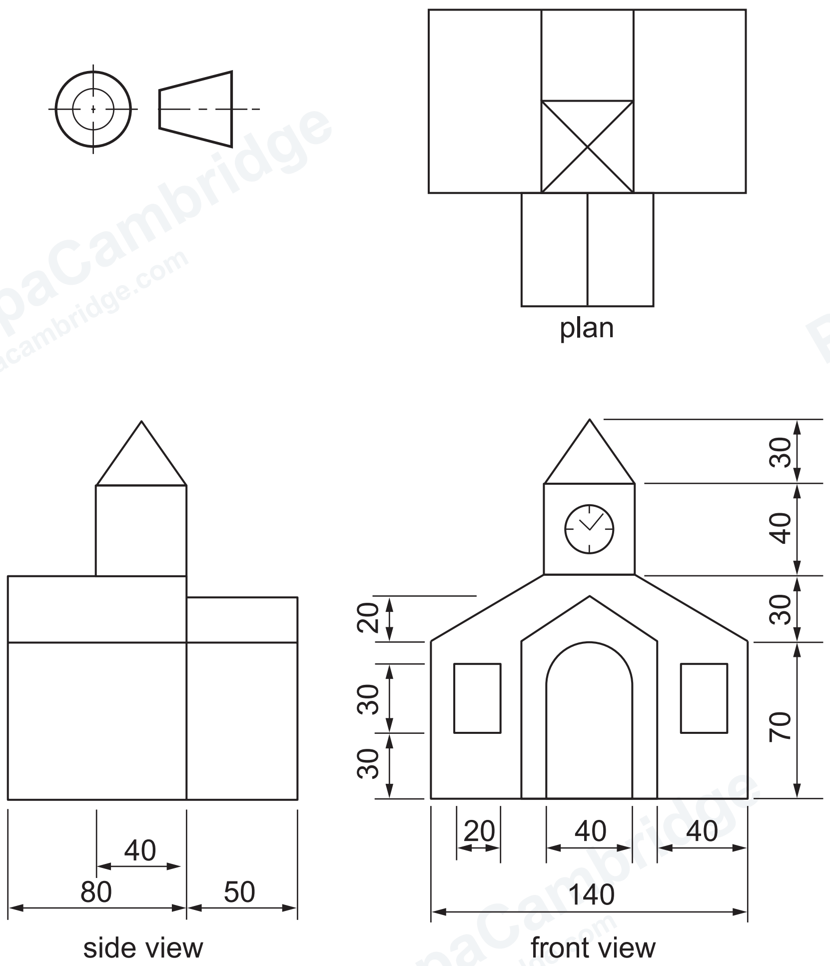
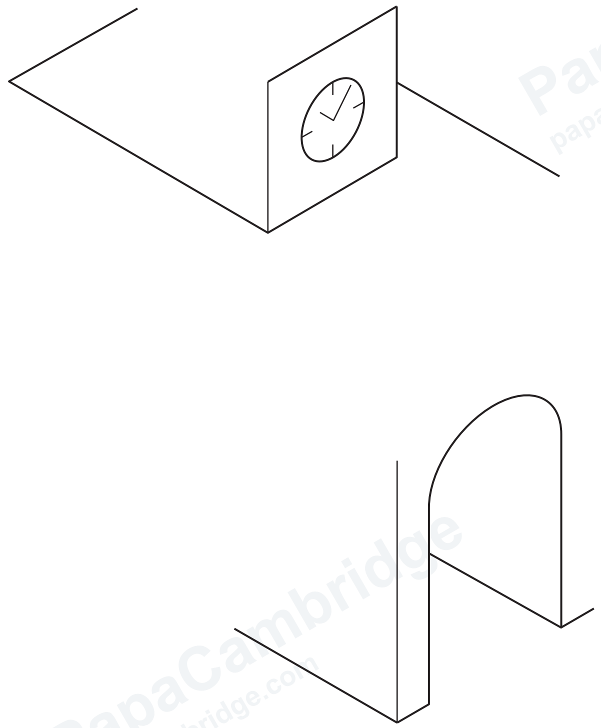
B5 Orthographic views of a model train station are shown below.



(a) Complete the isometric view of the model train station to a scale of 1 : 2. [13]



isometric view

(b) The model train station will be made from thin card and assembled by the user.

The developments (nets) for the models will be produced in quantities of 5000.

(i) State a suitable method of printing the developments (nets).

.... [1]

(ii) State a suitable method of cutting and creasing the developments (nets).

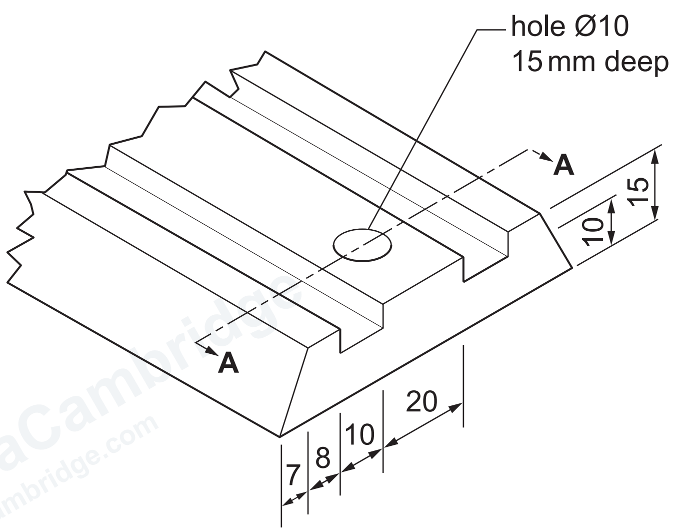
.... [1]

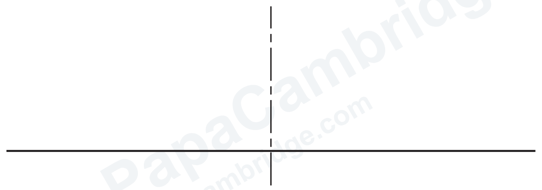
(c) The clock tower of the model train station will be made from two blocks of Styrofoam.

Complete the table below to show **one** tool / item of equipment for each process used in the making of the clock tower. [2]

process	tool/item of equipment
cutting out the Styrofoam blocks	
smoothing off the edges	glass paper
fixing the blocks together	

(d) Part of a length of toy train track is shown below.

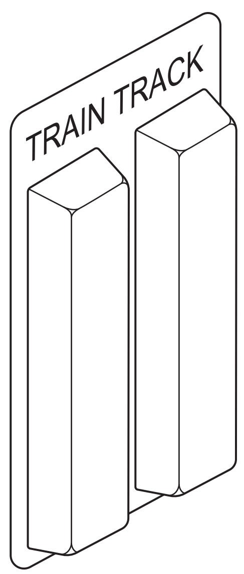




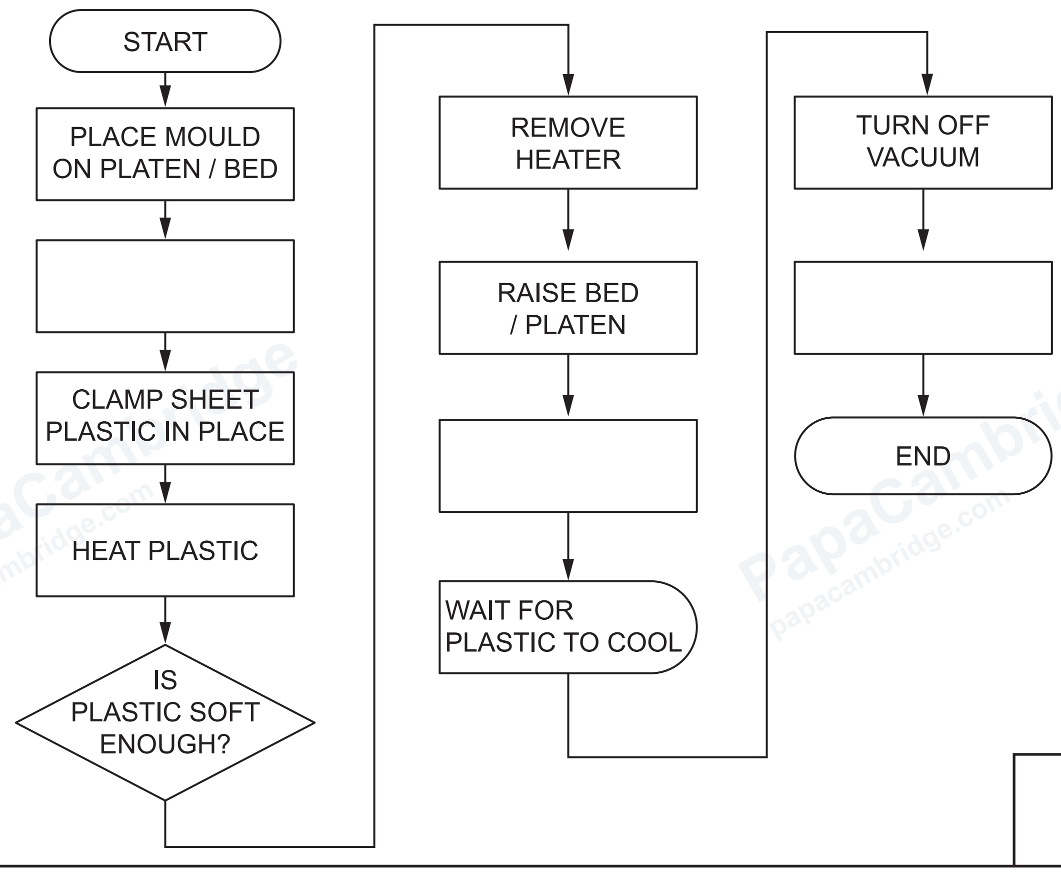
sectional view **A-A**

Complete the full-size sectional view **A-A** through the toy train track. [4]

(e) A vacuum formed blister pack for the toy train track is shown.



Complete the flow chart to show the vacuum forming process. [4]



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